

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The conversation emphasizes the need for more innovative and secure key management systems for crypto assets. It explores various solutions, including MPC for institutional use and AI for consumer-level applications. The importance of user literacy and proper education in cybersecurity is also highlighted, with a focus on risk assessment and secure practices. Future innovations, such as integrating AI to reduce human operation points, are discussed, and a three-year competition is proposed to drive innovation in key management.

Key Discussion Points

1. **Key Management Systems:**
 - The current state of key management is discussed, with a focus on the need for more innovative and secure methods.
 - Multi-Party Computation (MPC) is proposed as a potential solution for institutional use, while AI and delegate authority are suggested for consumer-level solutions.
2. **AI in Key Management:**
 - The role of AI in managing digital wallets is explored, with concerns about trustworthiness and the need for scoped access.
 - It is emphasized that AI should be a separate process from the key holder to ensure security.
3. **Risk Assessment and User Literacy:**
 - The importance of risk assessment and user literacy in cybersecurity is highlighted.
 - Operators and executives need proper education and training to manage digital assets securely.
 - There is a significant difference in the level of knowledge required for end-users versus institutional users.
4. **Future Innovations:**
 - The discussion includes brainstorming about future innovations in key management, such as integrating AI to reduce human operation points and improve risk management.

- A three-year competition is proposed to foster innovation in this area.

Action Items and Next Steps

1. **Develop MPC Solutions for Institutional Use:**
 - Create robust MPC solutions that can be used by institutions for managing digital assets securely.
2. **Integrate AI for Consumer-Level Key Management:**
 - Develop AI systems that can manage smaller wallets and delegate authority, ensuring scoped access and the ability to revoke access when necessary.
3. **Educate Operators and Executives:**
 - Provide proper training and education for operators and executives to ensure they understand the risks and best practices in key management.
4. **Foster Innovation Through Competitions:**
 - Organize a three-year competition to encourage innovation in key management systems, focusing on integrating AI and improving risk management.
5. **Promote User Literacy:**
 - Develop programs to improve user literacy in cybersecurity, especially for end-users who may have limited technical knowledge.
6. **Regularly Review and Update Practices:**
 - Continuously review and update key management practices to stay ahead of emerging threats and technologies.

Detailed Session Summary

The conversation revolves around the management of digital assets, particularly focusing on key management systems for crypto assets. The discussion delves into the challenges and potential solutions for secure key management, including the use of multi-party computation (MPC), artificial intelligence (AI), and user literacy in cybersecurity.

Appendix: Full Meeting Transcript (Anonymized)

58.6 --> 60.0: This is a meeting transcript.
60.0 --> 90.0: This is a meeting transcript.
95.2 --> 116.5: This is a meeting transcript.
121.4 --> 125.1: Thank you all. We'll get started with the next session.
133.1 --> 139.1: So to follow on from the previous session where we were discussing PQC and crypto agility,
139.4 --> 141.1: so more about the signatures,

142.0 --> 146.5: this session we're following more of the key management theme.

146.5 --> 152.7: So it does overlap more with governance, I guess, in this case.

153.3 --> 158.1: Things such as zero-knowledge proof of authentication, multi-party computation,

158.8 --> 161.9: the key lifecycle management,

162.4 --> 175.1: and I actually think that quantum or Q-Day is going to be one of the most significant key lifecycle management moments in history.

175.1 --> 179.8: So this is what we're preparing for and this is what this session is about.

180.4 --> 193.5: It's about building a game theoretic incentive loop for good key management in the PQC era and being prepared for that.

194.5 --> 204.8: So the key purpose of this session is to come up with some of the evaluation criteria for the competition and the benchmarks

204.8 --> 214.7: and some of the different aspects of key management that are going to be within the scope of the competition,

215.0 --> 220.7: but also to explore the innovations around key management.

220.7 --> 231.0: So bringing as many voices into the room with different ideas of PQC key management and the rotation and the lifecycle of that.

231.4 --> 242.3: So then we can aggregate and get towards a closer standard which will support with the overall Q-Day transition that is inevitable.

243.6 --> 245.7: So I'm going to pause there.

245.7 --> 249.7: Is there anything that I missed framing this session?

255.7 --> 256.8: Thank you.

256.8 --> 273.1: So I would like to add more context to why there is a sub-topic, a sub-category of the key management part inside NEDO's Japanese government competition.

274.1 --> 282.6: Last year, two years ago, DMM Bitcoin was attacked, compromised.

283.4 --> 288.4: The truly compromised part was the custodian company.

293.2 --> 296.8: DMM Bitcoin has a contract with that custodian company.

297.7 --> 300.9: Then that custodian company system was attacked.

301.9 --> 302.5: So that happened.

303.6 --> 315.0: Then that kind of blockchain custodian companies utilize that hardware security module as a system to manage the keys deposited.

316.3 --> 324.4: But they are usually utilizing multi-service system or multi-party competition key management system, something like that.

325.2 --> 328.3: But there is no standard for that kind of system.

329.3 --> 350.4: Many suppliers, many companies, many suppliers design their own multi-service system, their own multi-party computing system with their own way without any sufficient security evaluation or assessment.

351.4 --> 352.1: There is no standard.

353.1 --> 360.9: Creating a standard in the security is important to have

e a clear idea on how much risk remains.

362.0 --> 375.0: So with that clear system model and a clear standard, a clear risk assessment, we can limit the existence of the remaining risk.

375.0 --> 382.5: That is a huge mean for the operation of the company or risk mitigation purpose.

383.3 --> 387.4: So I strongly believe that we need to create a standard for key management system.

388.8 --> 397.0: Creating a standard, back to that Mitchell question, creating a standard is a good idea for the risk management purpose.

398.0 --> 405.8: At the same time, creating a standard will close the future chance to the innovation, might be.

406.6 --> 411.5: So this is the reason why utilizing competition is a great idea to create a standard.

412.1 --> 414.5: It's facilitating future innovation.

414.5 --> 423.9: So this is the reason why key management is added to that competition by Japanese government.

424.9 --> 430.4: We are thinking the PQC era risk management system.

431.4 --> 434.3: That is my additional explanation on that.

434.8 --> 440.5: Why the Japanese government started competition for the key management.

441.5 --> 460.5: So we've got a risk management theme around the keys, which then leads to better practices like insurance for these custodians that are holding assets on behalf of people and on behalf of exchanges and entities.

461.5 --> 475.2: And then we're also hosting a competition in order to allow a broad evaluation of the sort of key management systems that can become standards.

476.2 --> 487.5: I think one other observation that I've made in the market is a lot of the key management systems are very, very similar.

488.2 --> 497.8: But by establishing yourself as a key management system, just Fireblocks comes to mind.

497.8 --> 510.8: So Fireblocks established itself as an advanced key management system, MPC, and they captured a lot of the market in terms of digital asset custodians because of that.

511.9 --> 519.1: So, yeah, the question that comes to mind is, should the standard be made?

519.1 --> 522.9: Well, I think that the standard should be made in more of a public forum.

523.1 --> 533.9: So through the competition to create a more fair market, rather than sort of allowing for the private industry to capture the market and then push that as a standard.

533.9 --> 557.3: And I think this is the tradeoff that we're kind of evaluating all the time of whether or not it's best to have the free market determine the standards through just forces of like accretion over time, or we need to have a more open dialogue on the standards.

558.3 --> 567.2: I'm going to pause there. Any thoughts from the room with regards to the key management side of the competition?

575.3 --> 576.8: So there are a few different.

576.8 --> 590.2: I didn't ask that kind of thing to him before this session. I'm sorry. I would like to hear Takei-san's view about that.

590.2 --> 601.9: Of course, he's a part of the blockchain company in Japan, and of course he cannot say the detail about what happened inside the company.

601.9 --> 625.4: But can you explain something about what is the reality in that blockchain company, what kind of key management technology is used generally, and what kind of operations are done inside that company?

625.4 --> 639.0: I think it's better to have the current situation happening in the blockchain company. Can you share your thoughts on that?

644.0 --> 650.4: Hi, I'm Yuto. Actually, I think we will be speaking tomorrow in some session.

651.4 --> 660.5: Anyway, so I'm working for one of the Japanese crypto exchange, Merocoin.

660.5 --> 678.2: We, as custodians, have multiple pragmatic problems that we want to tackle.

678.2 --> 680.3: So problems to be solved, I mean.

680.8 --> 686.3: So the first one is, of course, technical perspective, like how the key should be secured.

686.7 --> 704.0: But the second one is, as Matsuo-sensei, like Professor Matsuo mentioned, operational perspective, especially day-to-day how we use the key, how to securely operate the signing procedures.

704.0 --> 714.9: And the third one is more like power management, power here meaning the company's authority, for example, like CEO should be able to access the key.

714.9 --> 721.4: Is it really feasible, like organizational structure?

722.0 --> 738.1: Or otherwise, are there multiple people who should be accessible to the key in case of emergency, like disasters or some other cases, like merger or M&A of companies, for example.

738.1 --> 748.3: So we have to think about how, here, there are multiple ways of solving these issues.

748.5 --> 763.2: Of course, like secret sharing, MPC, or physically split the key into different places, like one of the key shares should be offline, the other should be in the HSM, for example.

763.2 --> 776.3: So that's kind of a splitting the power, splitting the key share is actual design question of how the exchange should think about it.

777.1 --> 783.1: And I don't think there is kind of a standardized template answer to this question.

783.1 --> 792.0: For example, like one company may have like several C-class people who should have power over the key share.

792.4 --> 809.3: But maybe another company should like design, another company may design only the security information officer, like CISO should be able to access their master key and the other key should remain like automated, for example.

809.3 --> 830.0: So I don't have like a clear answer to this kind of des

ign goal, nor standardized goal, but it may be really interesting to think about what other options available to segregate power amongst those authorities within the one organization.

831.0 --> 849.4: And also like, well, what if one company like coordinates those kind of signing procedure with subprocessor, for example, like whether the custodians company, like external key vendors, for example, like Fireblocks or some other companies as well.

849.4 --> 860.0: What are the kind of right way to split the key shares amongst those subprocessors of the key share?

860.3 --> 862.3: What would be the threat model?

862.9 --> 867.2: So that's kind of a question that we day-to-day face in the crypto exchanges.

870.2 --> 872.1: That is a good answer.

872.1 --> 881.5: And I have that one thing we need to consider for the competition about operational feasibility.

883.8 --> 896.9: Even if that's a very good, mathematically good technology, but the operation is infeasible, so it will open another vulnerability for operation or something like that.

896.9 --> 903.4: So that operational feasibility is one of the important criteria for evaluation for the technology.

904.9 --> 912.3: And we need to consider the both, the normal time or emergency time, both.

914.1 --> 922.3: And at the same time, the easy explanation to JFSA is important.

923.1 --> 931.5: When some incident happens, they need to explain much things to JFSA from the supervision point of view.

931.5 --> 939.3: And easy to explain something to government or public is another important thing.

943.6 --> 956.0: So from what you said, I really like this three-tiered problem approach and framing, where there's the security, which typically happens at inception.

957.0 --> 960.5: And then, of course, you need to use the key.

961.0 --> 963.7: So there's signing, which is another vector.

964.0 --> 972.2: And then there are power dynamics as to who has access and at what time and with what permissions and protocol.

973.3 --> 975.9: So this framing, I enjoy a lot.

976.1 --> 977.2: So thanks for sharing that.

978.1 --> 999.5: One thing that came to mind is perhaps there should be almost like a key of keys in a way, where there's a key that goes through and actually outlines the governance within these three frames and environments.

1001.6 --> 1003.6: Just something that came to mind.

1004.4 --> 1008.6: Another point on that is the security side.

1009.2 --> 1019.1: It's almost like it shifts from hardware to software in that path as well, where you have the security is mostly evaluating the hardware of where it sits.

1019.6 --> 1026.2: And then the signing is like that connection between

the software, so the blockchain, for example, and the hardware.

1026.6 --> 1031.9: And then the power is mostly all software or the person and software.

1031.9 --> 1036.6: So it also has this track between the two.

1037.2 --> 1054.9: So when we do go towards the competition, perhaps that could be almost like a heat map or a pathway of these are the things that we'll be evaluating against is the initial security of the key ceremony.

1054.9 --> 1064.8: And then the signing aspect of how that do you need to go into a Faraday cage for particular key signatures or these sorts of things.

1065.4 --> 1071.7: And then the power dynamics gets way more into the governance side of things.

1072.5 --> 1075.3: So that was just my review.

1075.5 --> 1077.3: And yeah, thanks for your contribution.

1077.6 --> 1078.4: It was really great.

1086.8 --> 1086.9: Thanks.

1096.9 --> 1105.0: I guess we have some people from Ethereum Foundation who have knowledge on zero-knowledge proofs.

1105.0 --> 1115.8: Do you think zero-knowledge proofs are sort of beneficial here in terms of where do they sit on that three-tiered pathway?

1116.9 --> 1119.4: If I could call upon you guys.

1124.5 --> 1127.9: Like for authentication or for the signatures?

1135.0 --> 1146.2: ZKP is not the well-known technology to make MPC or key management right now.

1146.8 --> 1152.4: But I think that, as you pointed out, it's available in the future.

1153.4 --> 1175.3: And mainly in MPC schemes or key management schemes, instead of, for example, account obstruction, FH is well used.

1175.3 --> 1184.2: And some niche technology like garbled circuit is well used for MPC.

1185.4 --> 1196.6: And if you want to use ZKP for the key management system, maybe account obstruction will be better than MPC.

1199.2 --> 1203.1: There are two paths to make key management.

1204.0 --> 1208.7: If you want to use ZKP, maybe account obstruction is one of that.

1213.6 --> 1216.5: You referred to authentication.

1218.6 --> 1221.4: Authentication-wise, it's well used.

1221.5 --> 1235.1: Maybe ZKPath and using the passport information to make an identity which is not revealing a certain amount of information.

1235.7 --> 1237.8: We can use ZKP for that.

1238.8 --> 1250.0: And ZKP was avoided before simply because the client-side ZKP was not fast enough before.

1250.1 --> 1254.9: But right now it's totally available, even on browsers.

1254.9 --> 1270.4: So, I think, as a conclusion, before this year or last year, using ZKP for authentication in MPC or key management was much di

fficult.

1271.2 --> 1274.1: People didn't even try.

1274.5 --> 1277.5: But maybe it will be a better option after this.

1280.5 --> 1286.1: From my understanding, ZKP is not feasible right now, as Leono mentioned.

1286.9 --> 1294.1: However, in the threshold signature scheme, like Fire blocks supported,

1295.6 --> 1305.8: inside the threshold signature scheme, we are using ZKP to prove that this share came from master keys.

1306.8 --> 1313.2: So, partially, we can use ZKP, but it's not mainstream, I guess.

1325.8 --> 1326.8: Thank you.

1336.9 --> 1343.1: Did we have any other points on the zero-knowledge side of things for key management?

1346.7 --> 1351.8: One thing that came up when Leono was speaking was,

1353.9 --> 1362.7: there's going to be this combination of personhood credentials and key management, by the sounds of it.

1362.7 --> 1366.3: And the personhood credential side of things uses the zero-knowledge proof.

1366.5 --> 1372.8: And it could almost add an extra layer of protection there, with some of these cases that have been discussed.

1373.4 --> 1378.1: But we need to, of course, ensure that the personhood credential is secure.

1378.3 --> 1385.9: So, we should use zero-knowledge proofs for that, if we're going to be constantly calling that credential in order to make signatures.

1386.9 --> 1391.7: There are a few different approaches to personhood credentials,

1391.9 --> 1398.2: and we'll be discussing that in the session after lunch, the proof of personhood.

1398.7 --> 1406.1: So, I would like to use the research from this conversation in this group,

1406.4 --> 1412.2: and the research from the personhood conversation together for future Begin works.

1412.2 --> 1416.2: I think that that is a good initiative to move forward,

1416.8 --> 1426.0: because I think there's going to be a great overlap there around what credential, when, which credential,

1426.2 --> 1431.0: how many do you need to present in order to do signatures.

1431.7 --> 1433.2: And I think we could also potentially...

1434.2 --> 1441.4: I'm really enjoying this secure signing power dynamic triplet.

1441.8 --> 1447.0: So, perhaps we could map the different personhood credentials,

1447.3 --> 1453.0: or the different authentication requirements for each of these layers.

1453.9 --> 1461.5: So, of course, the first secure hardware layer would probably be the most robust requirement.

1461.5 --> 1467.1: Most likely have to have physicality or something along those lines.

1467.5 --> 1472.9: The signing maybe requires multiple personhood credentials,

1473.3 --> 1481.9: one being a biometric, one being a top-down one, like a bureaucratic one from the organization.

1483.0 --> 1490.8: And then for the power, it's almost like the personhood credential is used as proof

1491.5 --> 1493.6: or as like a record.

1494.2 --> 1498.8: So, if we do have events, insurance is required,

1499.1 --> 1503.0: there is some level of audit trail attached to accountability,

1503.9 --> 1507.3: because that's also something that I've seen in the market happen,

1507.6 --> 1509.6: where when an exchange gets hacked,

1510.5 --> 1515.8: the person within the exchange who is responsible is very hard to determine,

1516.8 --> 1523.9: because there is levels of privacy and security into that aspect,

1524.1 --> 1528.6: but also it's hard to find that accountability

1528.6 --> 1536.3: if you don't have standard procedures for identifying that.

1538.7 --> 1543.9: And can I ask one question to add some new angle?

1543.9 --> 1551.1: So, as I mentioned at the first part of this session about the background,

1551.5 --> 1556.6: so why the Japanese government is thinking to add key management for the competition,

1556.8 --> 1558.5: is it just for the custodian system?

1559.7 --> 1563.8: Obviously, the data signatures or keys inside,

1564.9 --> 1567.9: general people deposit it to the custodian system,

1568.1 --> 1570.4: that key itself is very important, obviously.

1570.4 --> 1576.1: So, this is the reason why we have multi-city or something like that for the custodian system.

1577.0 --> 1580.4: But, you know, originally,

1581.5 --> 1587.1: that attack surface for that incident was some, you know,

1588.5 --> 1590.3: not the key management system,

1592.2 --> 1598.8: it's a persistent, kind of a persistent threat,

1598.8 --> 1603.5: you know, receiving message over the LinkedIn,

1604.2 --> 1606.4: it was about something, this is the attack surface,

1606.6 --> 1609.5: the true attack surface is the start of that attack.

1611.7 --> 1614.8: You know, we're thinking that kind of thing, you know.

1615.7 --> 1620.0: There are many different types of keys used in the current Ethereum-based system,

1620.5 --> 1624.8: you know, including, you know, smart contract,

1625.1 --> 1629.8: other types of, there are many services on the top of that Ethereum blockchain.

1630.6 --> 1637.8: So we, you know, I cannot imagine how many different types of keys are used in that entire ecosystem.

1639.1 --> 1644.5: Of course, protecting layer one, you know, signing key for the transaction is important,

1645.1 --> 1649.5: but we might have another type of key, should be protected at the same level,

1650.6 --> 1651.8: and I cannot imagine.

1651.8 --> 1658.0: But we need to restart the other keys to be protected,

1658.6 --> 1661.3: or to have some standard model of the operation.

1662.9 --> 1667.6: I don't know, but when I thought that, you know,

1667.8 --> 1674.5: that kind of layer one signature, signing key attack happened so often,

1675.8 --> 1678.4: of course that, you know, coin check incident is the same,

1678.4 --> 1688.9: but the coin check case was on, it's January, February of 2018,

1690.3 --> 1691.4: eight years ago.

1692.3 --> 1696.6: At that moment, most use cases are just Bitcoin or, you know, layer one coin.

1697.2 --> 1703.6: There's no complicated smart contract system at that moment.

1703.6 --> 1709.5: But there's no layer two protocol, including Lightning Network.

1709.6 --> 1712.4: Lightning Network was invented in 2015, I think.

1713.2 --> 1720.1: But in 2018, still, Lightning Network is not active.

1720.9 --> 1726.2: But now we should care about what kind of key are used in the layer two,

1726.3 --> 1727.4: including Lightning Network.

1728.4 --> 1733.2: So we should care that different types of keys should have that

1733.2 --> 1735.6: at the same level of protection mechanism.

1736.5 --> 1740.6: But I have no idea what kind of key are used in this ecosystem entirely.

1741.4 --> 1747.8: And if we have that other important key to be protected,

1748.5 --> 1751.1: do we use multi-signature for that?

1751.2 --> 1751.9: I don't think so.

1751.9 --> 1757.1: But we need to restart that key should be protected in the current situation.

1766.0 --> 1770.2: Maybe I can add some kind of insights on top of it.

1770.8 --> 1775.3: So, especially, like, even now, like in exchanges,

1775.5 --> 1779.9: we do have, like, of course, a root key, like for the root wallet

1779.9 --> 1781.8: of the customer's funds, of course.

1782.3 --> 1787.1: But not only that, we do have, for example, like stak

ings, validators keys,

1787.3 --> 1793.9: or we do have, like, smart contract key for a smart contract wallet,

1795.1 --> 1798.3: or some other use cases as well.

1799.2 --> 1802.3: For example, like, in the future, we may provide some dApps,

1802.3 --> 1810.2: like, use cases for the customers or some other purposes, applications.

1810.9 --> 1816.4: In such cases, we may need to do some kind of key ceremony

1816.4 --> 1821.7: for those kind of decentralized apps for each scenario.

1822.0 --> 1823.7: Like, is it really feasible or not?

1823.8 --> 1826.0: That's really kind of interesting question.

1826.0 --> 1835.5: And, of course, at the exchange level, we need to visualize the, like,

1836.2 --> 1838.0: power dynamics that are mentioned earlier.

1838.2 --> 1842.6: Like, which key interact with which other keys, for example.

1842.9 --> 1844.7: So, maybe, like, validators funds.

1845.4 --> 1848.5: At first, like, validators key may have no value

1849.0 --> 1855.0: unless there is no staked fund on top of those kind of validators assets.

1856.5 --> 1863.3: So, but if we move some of the fund from the master wallet to the validation wallet,

1863.6 --> 1874.3: in such a case, like, the fund, like, value of the key virtually migrate from one key to the other.

1874.7 --> 1878.4: So, that's kind of, like, the change of the value,

1878.6 --> 1882.9: like, transition of the value from one key to other happens day to day.

1882.9 --> 1888.3: So, and we also need to understand, like, at any time,

1888.7 --> 1892.8: that which key is managed by which group in the company,

1893.0 --> 1899.0: like, which organization or team, I mean, like, within the exchanges, for example.

1899.4 --> 1907.0: And what are the operational, like, procedures to, for example, like, rescue the key,

1907.1 --> 1911.0: or I mean, like, kind of use a backup key in case of emergency.

1911.0 --> 1916.0: So, that really makes the situation complicated in the exchanges.

1916.5 --> 1921.9: So, like, simply say, like, simply put, we, as an exchange,

1922.0 --> 1928.2: don't want to make, don't want to, like, add another complexity

1928.2 --> 1936.1: or add another layer to the key management at the entire system on our own.

1936.1 --> 1941.0: So, if we can make everything simple, as, like, clear enough,
1941.6 --> 1945.9: that's really good for our operational perspective.
1946.4 --> 1954.1: And also, it's really easy to make explanation, I mean, justification for regulatory bodies
1955.8 --> 1962.4: that's, well, to explain how we operate the customer's funds to the regulators.
1962.4 --> 1968.9: So, well, to wrap up, I mean, like, we, as an exchange,
1969.1 --> 1972.7: don't want to add any complexity further.
1973.4 --> 1977.2: Like, if we can make everything simple, like, keep it simple,
1977.3 --> 1979.1: then that's really the best cases.
1979.4 --> 1984.3: But at the same time, we want to, like, well, also, like, understand
1984.3 --> 1989.3: what other, like, additional layers of, like, additional, like, applications
1989.3 --> 1993.9: when there is, like, new blockchain applications are developed.
1994.2 --> 1998.5: So, that's kind of a tricky perspective in our operation.
2012.8 --> 2018.5: So, last week, ETH Foundation announced the next generation's roadmap,
2018.9 --> 2024.3: and they announced that we're using the native account abstraction.
2024.3 --> 2030.0: So, in the future, we don't need any EOA account.
2030.6 --> 2037.2: So, I guess, as well as exchange, you support the account abstraction, right?
2037.8 --> 2038.6: Is this correct?
2041.8 --> 2049.5: So, I think it's hard operation for any other institution and company, I guess.
2051.1 --> 2052.4: Yeah, that's it.
2052.4 --> 2052.6: Yeah.
2055.6 --> 2056.7: One question.
2056.9 --> 2064.8: So, the scope of the key management competition will be basically for exchanges
2064.8 --> 2067.3: and custodial use cases, is that correct?
2073.0 --> 2074.4: Yes, I believe so.
2075.4 --> 2078.5: It's mostly focused on the custody situation.
2078.5 --> 2087.0: Do you think this competition is kind of scoping also cyber security?
2088.9 --> 2089.1: Yeah.
2089.3 --> 2100.2: Yeah. So, not only kind of theoretical crypto management, it's more like operation stuff, right?
2100.6 --> 2101.0: I see.
2102.3 --> 2103.1: Thanks.
2103.9 --> 2110.8: Yeah. And I think if I can tell what you're thinking,

2111.0 --> 2117.2: operations adds lots of layers of complexity more than the theoretical.

2118.0 --> 2121.3: Yeah, of course. But it's more practical, of course.

2121.9 --> 2128.7: What I want to think is like, no, it's also for post-quantum competition as well.

2128.7 --> 2132.9: I think we should define the advisory.

2134.1 --> 2140.0: If we can define the advisory, the participant can make a good system.

2141.1 --> 2147.7: Basically, no, it's also done by academic staff in cryptography,

2148.1 --> 2154.7: and also it's quite common in cyber security area.

2155.7 --> 2164.7: So, yeah, defining advisory can be one of the steps to define the scope of the competition in general.

2166.4 --> 2168.3: Yeah, I think that's a great steer.

2168.7 --> 2174.7: And it also aligns with the previous conversation around the short-term and long-term risk,

2175.7 --> 2182.8: and that defining the adversary or the attack is going to be helpful for us.

2192.4 --> 2196.1: When we start talking about cyber security, it's a bit broad,

2196.8 --> 2199.7: and there are too many huge coverages.

2201.0 --> 2206.1: So, I should talk that most of them might be out of scope of the competition.

2207.1 --> 2212.1: But if we can have some assumptions,

2212.1 --> 2216.5: so this is a point we should consider from cyber security point of view also.

2216.9 --> 2221.9: We can omit those points for the consideration.

2222.3 --> 2222.8: That is great.

2223.2 --> 2226.5: So, this is the reason why I mentioned that operational feasibility.

2227.2 --> 2230.9: So, if that's so, we can limit the points.

2232.0 --> 2235.1: So, the human should be in the loop.

2236.2 --> 2241.5: So, the number of points is related to that number of attack surfaces.

2242.1 --> 2247.1: So, if we can limit the points of the human should be in the loop,

2247.8 --> 2253.9: that is one important criteria to evaluate that technology.

2262.0 --> 2270.1: So, in the comparison between account abstraction and MPC solutions,

2270.1 --> 2276.7: at what point is the human in the room currently?

2277.9 --> 2282.9: Is it at the initial stage and then on signing?

2285.2 --> 2287.0: Is there a difference between the two?

2288.3 --> 2290.8: That's a question to the room.

2295.2 --> 2302.8: Is there a difference between when the human needs to be in the room for MPC versus account abstraction?

2303.6 --> 2304.8: Or is it the same?

2305.4 --> 2314.3: And if it's the same, then we can come up with a standard for managing that risk of the person.

2314.5 --> 2318.5: If it's different, then it might be a little bit more complex.

2318.5 --> 2321.8: But we could also use that difference as a comparison.

2333.3 --> 2336.9: To that question, from exchange perspective,

2337.2 --> 2340.5: we actually think about both the key ceremony,

2340.5 --> 2343.5: very initial stage of the system,

2344.8 --> 2349.6: and also as well as initial stage, operational stage as well.

2349.9 --> 2353.9: Because if we cannot guarantee the sanity of the system,

2354.1 --> 2357.0: the key is properly generated, for example,

2357.4 --> 2361.5: or key shares are properly distributed,

2362.9 --> 2366.3: for example, DKG protocols or any other protocol is fine.

2366.3 --> 2373.3: But we need to assure that there is no adversary in the stakeholders

2373.3 --> 2375.7: in case of MPC protocol, for example.

2376.4 --> 2380.5: So what's interesting about MPC, for example,

2380.7 --> 2385.6: is we need to only trust the protocol

2386.3 --> 2391.2: if the MPC protocol itself is properly designed.

2391.2 --> 2394.4: But at the same time, we use certain software

2394.4 --> 2398.1: to do those kind of key share distribution.

2398.4 --> 2401.5: For example, we may use third-party software

2401.5 --> 2406.3: to actually digitally sign, digitally use the key,

2406.7 --> 2410.1: I mean, key shares to generate a single signature

2410.1 --> 2413.0: after the multiple signing rounds.

2413.4 --> 2416.8: In such cases, we need to know if the software

2416.8 --> 2422.7: to support those kind of complex signing protocol

2422.7 --> 2424.8: is properly implemented.

2425.3 --> 2429.5: Or in the worst cases, maybe those kind of signing software

2430.5 --> 2432.6: is injected with some fault,

2432.8 --> 2436.3: I mean, to steal some bit from the key share, for example.

2436.6 --> 2439.2: That's actually one of my earlier work

2439.2 --> 2440.8: about what are cryptographic applications.

2441.8 --> 2445.1: But anyway, so from the exchange perspective,

2445.1 --> 2450.9: we need to always guarantee the key or key share

2450.9 --> 2457.1: is properly generated from the regular entropy.

2457.7 --> 2460.5: And also, those key share or key

2461.1 --> 2464.5: should be intact within those kind of premises.

2464.8 --> 2466.5: I don't know, like hardware security

2466.5 --> 2471.4: or any other way of storing the key or key shares as well.

2471.4 --> 2475.3: And when we use the key or key share

2475.3 --> 2479.0: at the point of using those key or key share,
2479.4 --> 2482.9: the software that touches those kind of key or key share
2483.4 --> 2487.6: should also be, I mean, genuine,
2488.0 --> 2490.0: I mean, not modified or injected
2490.0 --> 2495.4: with some adversarial, like malware or anything.
2496.8 --> 2499.5: So that's the kind of perspective
2499.5 --> 2505.5: we really think important from institutional perspective.
2513.8 --> 2515.1: So I've just written down,
2515.2 --> 2517.6: we should have a proof for the ceremony,
2517.8 --> 2519.7: we should have a proof for the software
2519.7 --> 2522.8: and we should have a proof for the hardware if that is.
2522.8 --> 2525.6: And if within the standard,
2526.3 --> 2529.8: we can outline like what type of proofs
2530.8 --> 2535.1: would be beneficial for saying that the hardware is working
2535.1 --> 2539.2: or that could also help with the order
2539.2 --> 2542.8: and with the regulatory enforcement
2542.8 --> 2544.7: and supervision side of things,
2545.2 --> 2547.2: because you don't necessarily,
2547.4 --> 2549.5: because one of the conflicts that comes to mind
2549.5 --> 2554.5: is the more information you reveal
2555.2 --> 2557.0: about your key ceremony,
2557.6 --> 2560.2: the less secure it becomes over time
2560.2 --> 2565.1: because people are able to work out the process
2565.1 --> 2568.4: and then find the gaps or find the vulnerabilities.
2569.2 --> 2573.2: So we should also be cognizant of that,
2573.3 --> 2576.1: that sometimes less information is more
2576.1 --> 2578.5: and that's the beauty of zero knowledge proofs, right?
2579.5 --> 2583.3: So perhaps there's a pathway here
2583.3 --> 2587.5: or an angle that we outline
2588.2 --> 2590.4: at what points of the ceremony
2590.4 --> 2593.5: and use of the key should a proof exist.
2594.9 --> 2599.5: And that could help with the, I don't know,
2599.6 --> 2604.4: the transparency whilst getting security,
2605.1 --> 2607.3: but also having a standard
2607.3 --> 2609.7: across all of the participants
2610.3 --> 2611.4: where it's very clear
2612.1 --> 2614.6: that once you do the ceremony, you create a proof.
2615.0 --> 2618.7: Once you work out the software that you're using,
2619.3 --> 2619.9: you do a proof.
2621.9 --> 2624.4: So yeah, or you could also prove
2624.4 --> 2628.4: that the MPC system is actually separate as well.
2628.5 --> 2630.8: Like you can already do those proofs,

2630.9 --> 2633.1: but maybe part of the standard
2633.1 --> 2636.0: that we develop through the competition
2636.0 --> 2639.2: outlines the enforcement of that.
2656.0 --> 2658.3: How about fund management?
2659.0 --> 2665.1: So when it comes to proposing a system for exchanges
2666.0 --> 2668.5: and custodians, entities,
2669.9 --> 2674.0: it's also fund management are related to that.
2674.2 --> 2678.3: For example, there's a hot wallet and cold wallet
2679.0 --> 2682.3: and money flows in cold wallet basically
2682.3 --> 2685.3: and occasionally they spend money to hot wallet
2685.3 --> 2687.1: or something like that.
2689.0 --> 2693.2: So when participants propose something
2693.2 --> 2697.2: for the key management competition,
2698.8 --> 2704.6: is that including the fund management as well or not?
2705.6 --> 2707.7: It just occurred to me as a question.
2718.7 --> 2721.3: A question to you, for example.
2721.3 --> 2729.0: So do you want a proposal including fund management
2729.0 --> 2732.3: like hot wallet, cold wallet structure
2732.8 --> 2735.3: and the key management related to that
2735.3 --> 2739.3: or do you just want a key management proposal
2740.2 --> 2741.5: in this competition?
2743.0 --> 2746.0: To be honest, from an exchange perspective,
2746.2 --> 2747.5: both are really important
2747.5 --> 2750.5: because right now in Japanese law states,
2751.6 --> 2754.1: I mean dictates, maybe everyone knows,
2754.3 --> 2758.0: but 95% of the funds should be in cold wallet
2758.0 --> 2763.1: and we need to follow that rule from exchange perspective.
2763.7 --> 2767.0: So sometimes we need to migrate the fund
2767.0 --> 2769.0: from the cold wallet to the hot wallet
2769.0 --> 2770.7: or vice versa as well.
2771.0 --> 2772.1: So that's really tricky
2772.1 --> 2776.1: because both funds are managed by us.
2776.1 --> 2777.8: I mean exchange side,
2777.9 --> 2779.0: but at the same time,
2779.1 --> 2781.1: in order to move the fund,
2781.2 --> 2783.7: we need to digitally sign to transition
2783.7 --> 2787.5: or transfer the fund from one wallet to another.
2788.0 --> 2790.1: If we can skip that process,
2790.6 --> 2794.1: I mean using the key is always dangerous, we think.
2794.7 --> 2799.1: So if we can automate those kind of fund management,
2799.3 --> 2802.0: that's really convenient for us as well.
2802.2 --> 2804.7: So in such cases, in such a sense,
2804.7 --> 2806.8: I mean the fund management proposal,
2807.1 --> 2810.6: that's really interesting for us
2810.6 --> 2814.3: and also like, well, sometimes.
2815.5 --> 2818.9: I have much fundamental question

2819.7 --> 2823.6: because in these several years in Japan,
2824.7 --> 2827.8: as you told that Japanese regulation requests
2828.7 --> 2832.7: exchange to hold 90% of entire asset
2832.8 --> 2836.5: in the so-called cold wallet.
2836.7 --> 2838.2: I don't know the definition,
2839.2 --> 2841.3: but it's a very fundamental question.
2843.0 --> 2845.9: There's no unified definition of the cold wallet,
2846.0 --> 2846.8: hot wallet, warm wallet.
2846.9 --> 2847.3: I don't know.
2849.7 --> 2851.9: It's a discussion of operational feasibility.
2852.1 --> 2854.9: Is it feasible idea to have the hot wallet and cold w
allet
2854.9 --> 2855.8: or segregation?
2858.3 --> 2861.8: I guess it's an entirely new model
2862.8 --> 2865.5: of protecting customer asset
2865.5 --> 2867.0: other than the hot wallet.
2867.4 --> 2868.5: What kind of segregation?
2868.7 --> 2869.8: It's very easy to understand.
2870.1 --> 2872.4: Traditional financial guides,
2872.7 --> 2877.0: like very strong physical vault system
2877.0 --> 2884.2: and some money in front of bank branch or something l
ike that.
2884.4 --> 2885.6: It's the same idea.
2886.6 --> 2890.6: But for blockchain world and the cryptography world,
2890.6 --> 2894.3: the idea of the hot, segregation of the hot wallet
2894.3 --> 2895.4: and the cold wallet,
2895.5 --> 2897.1: we should have that more,
2898.6 --> 2901.6: a better concept for that.
2902.5 --> 2903.6: Because in that,
2904.0 --> 2905.3: it's a possible discussion,
2905.7 --> 2911.5: but it's a very traditional idea.
2914.7 --> 2919.7: A part of the reason why the DMM incident happened
2919.7 --> 2921.9: or why it happened
2922.7 --> 2926.0: is the transition between the hot wallet and cold wal
let.
2926.5 --> 2927.9: That kind of human operation,
2928.3 --> 2930.0: human in the loop of things.
2931.0 --> 2935.0: If we can invent a very new model
2935.7 --> 2938.1: of asset management
2938.7 --> 2942.8: with reducing the point where the human is in the loo
p,
2943.7 --> 2948.0: that is a truly important topic for the competition.
2948.0 --> 2952.0: Of course, I don't know the JFSA will follow that res
ult.
2953.3 --> 2956.1: But if we can create that innovative way
2956.1 --> 2957.8: for the asset management,
2958.1 --> 2959.3: that is a huge win for us.

2959.7 --> 2962.2: And we can propose that kind of technology to JFSA
2962.2 --> 2964.2: to improve their regulation.
2965.6 --> 2968.1: I was a part of the working group members.
2969.9 --> 2972.6: It's just only my current opinion.
2972.9 --> 2975.5: It's not an opinion of that working group.
2975.5 --> 2980.2: But segregation between the hot wallet and cold wallet
2980.2 --> 2981.4: is a very bad idea.
2988.2 --> 2990.8: I mean, to some point,
2991.1 --> 2993.9: I understand what people who suggested cold wallet.
2994.2 --> 2996.5: But people from the PKI industry,
2997.4 --> 2999.2: like certificate authority,
2999.5 --> 3001.5: like X509 people,
3002.6 --> 3006.7: they operate those kind of root CA in the offline manner.
3006.9 --> 3007.6: I mean, area network,
3008.7 --> 3010.8: like completely offline basis.
3011.1 --> 3013.7: And they bring the CSR,
3013.8 --> 3015.5: like certificate signing requests
3015.5 --> 3018.7: from the online room to the offline CA room
3018.7 --> 3021.0: with metal detectors
3021.6 --> 3024.6: or any other detection systems in place.
3026.0 --> 3028.2: They want to keep those kind of offline
3028.2 --> 3030.2: like top key,
3031.6 --> 3033.5: like root key to be safe
3034.2 --> 3037.8: from any online attack vectors.
3038.2 --> 3040.2: So that idea is totally understandable.
3041.4 --> 3042.3: At the same time,
3043.5 --> 3046.9: the definition between hot wallet, cold wallet,
3047.2 --> 3052.5: that's really used arbitrarily right now.
3052.7 --> 3055.9: Maybe people say warm wallet.
3056.3 --> 3057.4: And what is that?
3058.2 --> 3059.4: We really don't know about
3059.4 --> 3060.9: what's the kind of differences
3060.9 --> 3062.8: between those kind of layers as well.
3063.0 --> 3065.2: So it's a tricky discussion.
3065.7 --> 3066.6: But at the same time,
3068.2 --> 3070.2: as back to the fund management question,
3070.5 --> 3075.6: if we can like safely manage
3075.6 --> 3076.7: all those kind of funds
3076.7 --> 3081.5: in kind of like grade of,
3082.0 --> 3084.3: I don't know, security, if I say,
3084.9 --> 3087.1: that's really nice for us.
3088.2 --> 3090.5: To limit the kind of exposure
3090.5 --> 3094.1: from the external threat at any time.
3099.2 --> 3103.0: Do you think shifting away
3103.0 --> 3105.4: from hot and cold into offline online

3106.2 --> 3107.4: makes more sense?
3113.4 --> 3115.6: Tricky because of the MPC as well.
3115.7 --> 3115.8: Yeah.
3118.2 --> 3118.7: I don't know.
3119.6 --> 3121.7: I don't have like right answer,
3121.9 --> 3124.6: well, answer to that question right away.
3129.5 --> 3132.0: Hot, cold, or offline online
3132.0 --> 3134.2: has the same meaning for JFS person.
3135.0 --> 3135.6: Yeah, that's right.
3135.8 --> 3136.8: Connected to the internet or not.
3137.1 --> 3138.3: Well, that's what I like.
3138.6 --> 3141.5: But if they're logically connected to something
3141.5 --> 3143.5: or logically disconnected to something,
3143.5 --> 3145.8: that other meaning,
3146.1 --> 3148.1: that makes sense for engineers
3148.1 --> 3149.4: or computer scientists.
3150.0 --> 3151.5: But for JFS,
3152.7 --> 3155.2: it means that connected to internet or not.
3157.0 --> 3159.5: So we need to invent some better words
3159.5 --> 3160.7: to express that.
3162.3 --> 3164.9: The important thing is it's accessible
3165.5 --> 3168.5: or not accessible by adversary
3169.4 --> 3171.0: or something like that.
3172.0 --> 3174.0: But we need to invent new words
3174.0 --> 3175.5: to express that, you know,
3175.6 --> 3179.2: access control to some important information
3179.2 --> 3180.9: at some time.
3184.6 --> 3186.9: I work with those kind of
3186.9 --> 3188.4: shared operating system,
3188.9 --> 3190.4: but in my understanding,
3191.0 --> 3195.0: they meant to make that too slow
3195.0 --> 3198.5: because if the procedure is slow,
3198.5 --> 3202.9: somebody can stop those procedures.
3203.5 --> 3205.0: So if it were instantly,
3205.5 --> 3207.8: when we notice it might be too late.
3208.2 --> 3210.2: So in my understanding,
3210.5 --> 3213.2: they are meant to design to be slow
3213.2 --> 3215.4: so that, for example,
3215.6 --> 3217.4: if you are driving a car
3217.4 --> 3221.0: and you are driving in front of the school zone,
3221.4 --> 3223.1: you drive slowly, right?
3223.4 --> 3228.5: So if you are processing huge assets,
3228.5 --> 3231.3: or huge assets who have a value,
3231.8 --> 3234.4: they are meant to make slow
3234.4 --> 3235.6: for those procedures.
3236.4 --> 3239.0: I'm not sure that is suitable
3239.0 --> 3240.8: for blockchain ecosystem,

3241.3 --> 3244.3: but I'm not sure,
3244.9 --> 3246.3: I can't say we,
3246.4 --> 3247.9: but at least me,
3248.6 --> 3250.5: is thinking about designing system
3251.2 --> 3252.8: with those kind of thinking.
3253.3 --> 3256.3: So I'm not sure
3256.3 --> 3259.3: like offline is very suitable,
3260.2 --> 3262.3: or air gap system is suitable for blockchain,
3262.6 --> 3266.6: but it might be easiest way
3266.6 --> 3270.9: to guarantee slowness of the procedure, yeah.
3275.0 --> 3276.4: If I may, I'm,
3277.3 --> 3279.3: actually like was that's really important
3280.3 --> 3281.3: and interesting perspective,
3282.1 --> 3283.4: like temporal perspective,
3283.6 --> 3285.8: is it really quick?
3286.3 --> 3287.8: Like attack can be completed,
3288.1 --> 3289.4: or does it take time
3289.4 --> 3291.7: to like slow down their adversary?
3292.0 --> 3293.3: So that's really important question
3294.2 --> 3296.3: in terms of those kind of offline management.
3296.8 --> 3299.0: And in case of like CA operation,
3299.2 --> 3300.3: like PKI's industries,
3300.6 --> 3302.3: like the end result,
3302.9 --> 3304.8: I mean end result here meaning
3304.8 --> 3308.5: like was after signing operation is done,
3308.8 --> 3310.5: that's most of the time
3310.5 --> 3312.6: like X.509 certificate,
3313.0 --> 3314.1: or CLR.
3314.1 --> 3317.5: So like was the format of the end result
3317.5 --> 3319.0: in the PKI industry
3319.0 --> 3321.5: is always easily validated,
3322.0 --> 3322.5: so to say.
3322.6 --> 3325.2: But in case of like digital asset industry,
3325.4 --> 3325.8: for example,
3326.2 --> 3327.8: maybe the signing outcome
3327.8 --> 3330.6: may be a transaction to a smart contract,
3330.8 --> 3332.3: and we don't really know
3332.3 --> 3334.2: what's the outcome
3334.2 --> 3337.0: of that transaction execution result is.
3337.2 --> 3337.6: For example,
3337.7 --> 3339.0: like well maybe that's a signing
3339.0 --> 3341.0: to the smart contract wallet,
3344.1 --> 3345.4: from one wallet to another.
3346.1 --> 3348.0: That end result
3348.0 --> 3349.4: is not really visible
3349.4 --> 3350.9: from the digital signature.
3351.7 --> 3352.9: While PKI industry,

3353.1 --> 3354.3: that's really clear
3354.3 --> 3356.4: because like certificate is signed.
3356.9 --> 3359.0: So that's only the certificate part.
3361.0 --> 3362.8: and we also like
3362.8 --> 3364.5: from digital asset perspective,
3365.3 --> 3367.4: when the key is compromised,
3368.1 --> 3369.0: most of the cases
3369.0 --> 3371.0: the attack is instantly done.
3371.2 --> 3372.8: So like if we notice
3374.1 --> 3375.4: that transaction is committed
3375.4 --> 3376.3: to the blockchain,
3376.5 --> 3378.1: that's already over.
3378.4 --> 3380.4: So if we found out
3380.4 --> 3381.8: if the key is compromised,
3383.7 --> 3386.0: we know that incident
3386.0 --> 3388.0: out of the blockchain monitoring,
3388.5 --> 3390.3: and in that sense,
3391.0 --> 3392.1: the fund is already
3392.1 --> 3393.6: drained from the wallet,
3393.9 --> 3395.9: and we really have nothing
3395.9 --> 3397.7: that we can do after that.
3398.1 --> 3402.2: So it's really important for us
3402.2 --> 3403.8: to, of course,
3403.9 --> 3404.9: slow down the attacks.
3405.1 --> 3406.8: That's one important perspective.
3407.5 --> 3408.3: The other is,
3408.9 --> 3409.3: of course,
3409.6 --> 3413.0: we let that never occur.
3413.2 --> 3414.7: I mean like in the first place.
3414.9 --> 3416.1: That's also like kind of
3417.4 --> 3418.5: way of thinking
3418.5 --> 3420.0: in exchanges as well.
3420.7 --> 3422.5: I like this analogy
3422.5 --> 3424.6: of driving through a school zone.
3424.9 --> 3425.4: I like that.
3426.2 --> 3428.3: One word that came to mind is
3428.3 --> 3429.9: perhaps it's
3432.4 --> 3433.3: machine friction
3433.3 --> 3434.8: and machine friction
3434.8 --> 3436.3: versus human friction
3436.3 --> 3438.9: in the process.
3439.8 --> 3443.4: And when it's a lot of money,
3443.7 --> 3444.9: we should increase
3444.9 --> 3446.2: machine friction.
3447.9 --> 3449.2: In the
3450.2 --> 3451.5: other cases,
3452.0 --> 3453.1: we should maybe increase

3453.1 --> 3454.8: or reduce human friction.
3455.8 --> 3457.8: Perhaps that's a good way
3457.8 --> 3460.1: to understand the problem
3460.1 --> 3461.6: that gets us away
3461.6 --> 3463.3: from hot-cold dichotomy.
3475.2 --> 3477.1: You mean like committing
3477.1 --> 3478.1: the intent first
3478.1 --> 3480.6: and you have to have
3480.6 --> 3481.9: certain period of time
3481.9 --> 3484.1: before the transaction
3485.2 --> 3486.2: is actually committed?
3486.7 --> 3487.3: Something like that?
3487.3 --> 3488.6: Like in the
3489.3 --> 3490.4: ACH in the United States?
3499.9 --> 3501.9: It doesn't have to be human friction.
3502.2 --> 3503.6: It can just be
3504.3 --> 3505.0: machine, right?
3505.1 --> 3507.0: And then the analytics
3507.0 --> 3508.5: can actually walk out
3508.5 --> 3510.7: and veto the
3510.7 --> 3512.0: transaction intent.
3515.1 --> 3516.5: Yeah, so I like that
3516.5 --> 3517.7: intent word as well.
3518.2 --> 3518.9: And I think
3518.9 --> 3520.1: we can actually
3522.1 --> 3524.2: lean into intent
3524.2 --> 3526.4: because a lot of the
3529.6 --> 3531.8: like I see the intent-based internet
3531.8 --> 3533.6: being the emergent internet.
3534.6 --> 3536.2: So, yeah, perhaps
3536.2 --> 3537.9: that is a good approach to take
3537.9 --> 3539.3: is that we
3540.4 --> 3541.6: express intents
3542.0 --> 3542.6: and then
3542.6 --> 3544.5: there is some level of
3544.5 --> 3545.5: temporal delay
3546.0 --> 3547.0: or some level of
3547.0 --> 3548.0: machine friction
3548.0 --> 3549.5: that goes into place
3550.0 --> 3550.6: and then
3552.0 --> 3554.0: you have the transaction execute.
3555.3 --> 3557.4: What's interesting about that as well
3558.0 --> 3558.6: is
3558.6 --> 3559.9: that number
3559.9 --> 3561.3: that like the amount
3561.3 --> 3562.3: of friction
3563.0 --> 3566.0: is something that can be

3567.0 --> 3568.4: like varied
3569.0 --> 3570.4: based on the type of risk.
3570.4 --> 3571.9: And so
3571.9 --> 3572.7: the risk assessment
3573.4 --> 3574.8: can actually inform
3575.4 --> 3576.9: the to what extent
3576.9 --> 3578.4: friction is important.
3579.1 --> 3579.7: Yeah.
3585.3 --> 3587.2: So that's generally what happens
3587.2 --> 3588.5: in trial-fire space, right?
3589.0 --> 3590.4: You first register
3590.4 --> 3591.4: the payment intent
3591.4 --> 3592.5: and then
3592.5 --> 3594.2: there's a risk analytics
3594.2 --> 3594.8: going on.
3595.2 --> 3596.7: And if it's small amount
3596.7 --> 3597.5: it just go through.
3597.9 --> 3598.5: If it's not
3598.5 --> 3599.5: you get
3600.4 --> 3601.3: 3D secure
3601.3 --> 3601.8: or push
3601.8 --> 3603.1: or something like that.
3603.4 --> 3603.8: And then
3603.8 --> 3605.0: if that's
3605.0 --> 3606.2: above certain amount
3606.2 --> 3607.5: when I was
3607.5 --> 3608.4: I sent
3608.4 --> 3609.3: like
3611.0 --> 3612.4: \$15,000
3612.4 --> 3613.5: or something like that
3613.5 --> 3614.2: to the United States
3614.2 --> 3616.1: I was actually
3616.1 --> 3617.7: asked to come
3617.7 --> 3618.1: to the
3618.1 --> 3618.5: branch.
3622.7 --> 3623.7: Yeah, so
3623.7 --> 3625.6: those trial-fire
3625.6 --> 3627.0: actually has those things.
3627.6 --> 3628.1: I mean
3628.1 --> 3629.1: if it was like
3629.1 --> 3629.9: \$1,000
3629.9 --> 3631.7: I think it just went through.
3636.7 --> 3638.0: So you could
3638.0 --> 3639.7: yeah, use the word broadcasting
3639.7 --> 3640.7: as well here.
3641.2 --> 3642.8: So you're like broadcasting
3642.8 --> 3643.4: the intent

3643.4 --> 3644.3: and then
3644.3 --> 3644.9: you
3646.0 --> 3647.4: have some sort of procedure
3647.4 --> 3648.5: and process in the middle.
3649.3 --> 3650.0: I guess
3651.5 --> 3652.9: MEV comes to mind
3653.8 --> 3655.2: here in a way
3655.2 --> 3656.4: because if you're providing
3656.4 --> 3657.5: an intent and a broadcast
3657.5 --> 3658.9: and it is
3658.9 --> 3659.9: a transaction
3659.9 --> 3660.6: that
3660.6 --> 3662.1: can be front-run
3662.1 --> 3662.5: or
3662.5 --> 3663.0: can
3663.0 --> 3663.7: like
3663.7 --> 3665.6: it's a transaction of
3666.6 --> 3668.4: economic significance
3668.4 --> 3669.3: like you're
3669.3 --> 3670.5: planning on selling
3670.5 --> 3671.6: or something like that
3672.5 --> 3673.4: that may
3673.4 --> 3674.2: come into
3674.2 --> 3675.7: play here.
3676.4 --> 3677.7: But if it's just
3677.7 --> 3679.0: key rotation
3679.9 --> 3680.9: or
3681.5 --> 3683.1: exchange-based
3683.1 --> 3684.0: governance
3684.0 --> 3684.6: between
3685.5 --> 3686.7: offline-offline
3688.5 --> 3688.9: hot-cold
3689.5 --> 3689.6: then
3690.5 --> 3691.9: it may not be as
3692.5 --> 3693.4: much of an issue
3693.4 --> 3695.0: to include that like
3695.0 --> 3695.4: friction
3695.4 --> 3696.4: both
3696.4 --> 3697.5: temporarily and
3699.1 --> 3699.8: physically.
3706.7 --> 3708.6: I think it's kind of
3708.6 --> 3709.4: funny to say
3709.4 --> 3709.8: but like
3709.8 --> 3710.8: actually is the
3710.8 --> 3711.5: transaction
3711.5 --> 3712.9: in terms of
3712.9 --> 3713.7: digital asset

3715.5 --> 3715.9: itself is
3715.9 --> 3717.2: actually an intent
3717.2 --> 3717.8: from computer
3717.8 --> 3718.5: scientific
3718.5 --> 3719.4: perspective.
3719.7 --> 3720.1: I mean like
3720.1 --> 3721.0: that's a transaction
3721.0 --> 3722.1: originally I mean
3722.1 --> 3723.2: in Bitcoin cases
3723.9 --> 3724.8: I'm going to
3724.8 --> 3725.3: use this
3725.3 --> 3725.7: Bitcoin.
3726.5 --> 3726.9: This is
3726.9 --> 3727.7: here is my
3727.7 --> 3728.3: digital signature
3728.3 --> 3729.3: so this is my
3729.3 --> 3729.7: intent.
3729.9 --> 3730.6: Actually that's
3730.6 --> 3730.9: kind of
3730.9 --> 3732.0: equal
3732.0 --> 3733.1: relationship
3733.1 --> 3734.0: between like
3734.0 --> 3734.9: digital signatures
3734.9 --> 3736.0: I mean
3736.0 --> 3736.8: Bitcoin's
3736.8 --> 3737.2: transaction
3737.2 --> 3738.0: data structure
3738.0 --> 3738.9: and also
3738.9 --> 3740.2: like personal
3741.2 --> 3742.4: as a human
3742.4 --> 3742.9: intent
3742.9 --> 3744.1: to transfer
3744.1 --> 3744.6: the fund
3744.6 --> 3745.4: from one
3745.4 --> 3745.8: place to
3745.8 --> 3746.1: another.
3746.8 --> 3747.4: But now
3747.4 --> 3748.0: like because
3748.0 --> 3748.3: of like
3748.3 --> 3748.8: smart contract
3748.8 --> 3749.5: applications
3749.5 --> 3750.4: or any
3750.4 --> 3750.7: other
3750.7 --> 3751.2: like
3751.2 --> 3751.8: complicated
3751.8 --> 3753.0: I mean
3753.0 --> 3753.9: complex

3753.9 --> 3755.9: way of
3755.9 --> 3757.9: digital fund
3757.9 --> 3758.4: use
3758.4 --> 3759.3: that makes
3759.3 --> 3760.2: the situation
3760.2 --> 3760.7: further
3760.7 --> 3761.5: like
3761.5 --> 3763.1: complex
3763.1 --> 3764.3: and the
3764.3 --> 3764.8: intent
3764.8 --> 3766.2: here
3766.2 --> 3767.6: maybe like
3767.6 --> 3768.0: for example
3768.0 --> 3768.5: like well
3768.5 --> 3769.8: we as an
3769.8 --> 3770.5: exchange want
3770.5 --> 3771.6: to transfer
3771.6 --> 3772.0: the fund
3772.0 --> 3772.6: from cold
3772.6 --> 3773.2: wallet to
3773.2 --> 3773.9: hot wallet
3773.9 --> 3774.5: or vice
3774.5 --> 3774.8: versa.
3775.0 --> 3775.6: That may be
3775.6 --> 3776.2: an intent
3776.2 --> 3776.9: but at the
3776.9 --> 3777.4: same time
3777.4 --> 3778.3: that should
3778.3 --> 3779.0: be expressed
3779.0 --> 3780.4: as a
3780.4 --> 3780.9: transaction
3780.9 --> 3781.9: or is
3781.9 --> 3782.3: it like
3782.3 --> 3783.1: well or
3783.1 --> 3783.9: more
3784.6 --> 3785.7: other like
3785.7 --> 3786.1: verbally
3786.1 --> 3786.8: expressed
3786.8 --> 3787.5: way or
3787.5 --> 3789.1: and who
3789.1 --> 3789.8: interprets
3789.8 --> 3790.4: that kind
3790.4 --> 3790.9: of intent.
3791.1 --> 3791.5: That's
3793.2 --> 3794.0: kind of
3794.0 --> 3794.8: a border
3795.6 --> 3796.2: is really

3796.2 --> 3797.0: becoming
3797.0 --> 3797.9: big
3797.9 --> 3798.4: nowadays
3798.4 --> 3799.6: I think.
3800.8 --> 3801.6: If that
3801.6 --> 3802.3: makes sense
3802.3 --> 3802.6: I don't
3802.6 --> 3802.8: know.
3803.6 --> 3803.9: I think
3803.9 --> 3804.2: there could
3804.2 --> 3804.8: actually also
3804.8 --> 3805.2: be a
3805.2 --> 3805.7: separation
3805.7 --> 3806.4: between
3806.4 --> 3808.2: who proposes
3808.2 --> 3808.9: the intent
3808.9 --> 3810.0: and who
3810.0 --> 3810.7: executes
3810.7 --> 3811.3: as well
3811.3 --> 3811.9: which adds
3811.9 --> 3812.3: like a
3812.3 --> 3813.1: layer of
3813.1 --> 3813.6: security.
3813.6 --> 3814.6: Well
3815.2 --> 3816.4: maybe not
3816.4 --> 3816.8: security
3816.8 --> 3817.3: but just
3817.3 --> 3818.5: yeah
3818.5 --> 3819.0: it is
3819.0 --> 3819.4: kind of
3819.4 --> 3819.7: security
3819.7 --> 3820.1: having
3820.1 --> 3820.7: more than
3820.7 --> 3821.2: one.
3825.1 --> 3825.6: So
3825.6 --> 3827.2: what I
3827.2 --> 3827.7: suggested
3827.7 --> 3828.2: was that
3828.2 --> 3829.6: like in
3829.6 --> 3830.3: something like
3830.3 --> 3830.9: financial
3831.6 --> 3832.4: transactions
3832.4 --> 3832.8: I mean
3832.8 --> 3833.3: in the
3833.3 --> 3833.8: traditional
3833.8 --> 3835.0: Bitcoin
3835.0 --> 3835.6: something like

3835.6 --> 3836.3: that the
3836.3 --> 3837.5: intent is
3837.5 --> 3838.1: transaction
3838.1 --> 3838.4: right.
3840.3 --> 3841.0: What I
3841.0 --> 3841.4: suggested
3841.4 --> 3841.7: was
3843.0 --> 3843.0: maybe
3844.7 --> 3845.0: it's
3845.5 --> 3846.0: safer
3846.0 --> 3846.3: to
3847.0 --> 3847.3: separate
3847.3 --> 3847.8: them into
3847.8 --> 3849.0: at least
3849.0 --> 3849.6: two steps
3849.6 --> 3850.6: right.
3851.3 --> 3851.9: So that you
3851.9 --> 3852.6: have chance
3852.6 --> 3853.4: to intervene
3853.4 --> 3853.8: right.
3858.5 --> 3859.5: I think
3859.5 --> 3859.8: also
3859.8 --> 3860.4: that it's
3860.4 --> 3860.7: more
3860.7 --> 3861.3: possible
3861.3 --> 3861.9: if
3863.3 --> 3864.2: to
3864.2 --> 3864.6: create
3864.6 --> 3865.2: the
3865.2 --> 3865.8: separation
3865.8 --> 3866.9: if the
3866.9 --> 3867.3: value
3867.3 --> 3868.0: like if
3868.0 --> 3868.2: you're
3868.2 --> 3868.6: worried about
3868.6 --> 3868.8: the
3868.8 --> 3869.1: speed
3870.0 --> 3870.5: I
3870.5 --> 3871.2: part of
3871.2 --> 3871.3: the
3871.3 --> 3871.7: problem
3871.7 --> 3872.1: and I
3872.1 --> 3872.4: think this
3872.4 --> 3872.9: will be
3872.9 --> 3873.3: one of
3873.3 --> 3873.5: the most
3873.5 --> 3873.8: common

3873.8 --> 3874.3: critiques
3874.3 --> 3875.0: is
3875.0 --> 3875.4: slow.
3875.9 --> 3876.7: So why
3876.7 --> 3876.9: are we
3876.9 --> 3877.2: bringing
3877.2 --> 3877.5: a slow
3877.5 --> 3878.0: system
3878.0 --> 3878.3: into
3878.3 --> 3878.5: a
3878.5 --> 3878.9: computer
3878.9 --> 3879.3: system.
3891.2 --> 3891.8: I
3891.8 --> 3892.3: that
3892.3 --> 3892.7: can
3892.7 --> 3893.1: make
3893.1 --> 3893.7: a
3893.7 --> 3893.7: of
3893.7 --> 3893.8: but
3893.8 --> 3893.9: not
3893.9 --> 3894.2: the
3894.2 --> 3894.7: transaction
3894.7 --> 3895.9: it doesn't
3895.9 --> 3896.2: actually
3896.2 --> 3896.8: touch the
3896.8 --> 3897.5: key
3897.5 --> 3898.2: side of
3898.2 --> 3898.3: the
3898.3 --> 3898.7: system
3898.7 --> 3899.2: it
3899.2 --> 3899.7: observes
3899.7 --> 3899.9: the
3899.9 --> 3900.4: system
3900.4 --> 3901.2: provides
3901.2 --> 3901.4: a
3901.4 --> 3901.8: review
3901.8 --> 3902.6: provides
3902.6 --> 3903.2: some
3903.2 --> 3903.5: sort
3903.5 --> 3903.6: of
3903.6 --> 3904.1: audit
3904.6 --> 3904.9: yes
3904.9 --> 3905.4: you can
3905.4 --> 3905.5: then
3905.5 --> 3906.0: probably
3906.0 --> 3906.8: have
3906.8 --> 3907.0: a whole

3907.0 --> 3907.3: bunch
3907.3 --> 3907.6: of
3907.6 --> 3908.2: lawyers
3908.2 --> 3908.6: blaming
3908.6 --> 3908.8: the
3908.8 --> 3909.0: AI
3909.0 --> 3909.5: instead
3909.5 --> 3909.7: of
3909.7 --> 3909.8: the
3909.8 --> 3910.1: human
3910.1 --> 3910.6: if
3910.6 --> 3910.9: things
3910.9 --> 3911.0: go
3911.0 --> 3911.3: wrong
3911.3 --> 3911.6: but
3911.6 --> 3911.9: at
3914.1 --> 3914.7: least
3914.7 --> 3914.9: it
3914.9 --> 3915.5: creates
3915.5 --> 3916.1: a
3916.1 --> 3916.5: human
3916.5 --> 3916.9: readable
3916.9 --> 3917.8: understanding
3917.8 --> 3918.4: of
3918.9 --> 3919.2: the
3919.2 --> 3919.6: intent
3919.6 --> 3920.0: that was
3920.0 --> 3920.4: made
3920.4 --> 3920.9: to
3920.9 --> 3921.6: broadcast
3921.6 --> 3922.0: this
3922.0 --> 3922.6: transaction.
3924.6 --> 3925.8: That could
3925.8 --> 3926.2: be one
3926.2 --> 3926.6: way that
3926.6 --> 3927.3: you solve
3927.3 --> 3928.1: for that
3928.1 --> 3928.8: objection.
3930.3 --> 3931.6: That actually
3931.6 --> 3932.4: reminds me
3932.4 --> 3932.6: of
3932.6 --> 3935.4: when I'm
3935.4 --> 3935.7: using
3935.7 --> 3936.6: AI for
3936.6 --> 3937.0: coding
3937.0 --> 3937.6: or some
3937.6 --> 3937.9: other

3937.9 --> 3938.4: purposes
3938.4 --> 3940.1: the AI
3940.1 --> 3941.6: keeps
3941.6 --> 3942.5: pinging
3942.5 --> 3942.7: me.
3943.0 --> 3943.8: Can I
3943.8 --> 3944.3: write this
3944.3 --> 3944.6: file?
3944.8 --> 3945.0: Can I
3945.0 --> 3945.6: modify this
3945.6 --> 3945.9: file?
3948.3 --> 3950.0: Within this
3950.0 --> 3950.4: session
3950.4 --> 3951.0: this
3951.0 --> 3952.3: operation
3952.3 --> 3952.8: this
3952.8 --> 3953.5: directory
3953.5 --> 3955.0: and I
3955.0 --> 3955.3: can now
3955.3 --> 3955.5: see
3955.5 --> 3956.8: maybe
3956.8 --> 3959.0: like in
3959.0 --> 3959.9: custodians
3959.9 --> 3960.5: management
3960.5 --> 3961.0: I mean
3961.0 --> 3961.7: AI
3961.7 --> 3963.2: not the
3963.2 --> 3963.7: institutional
3963.7 --> 3964.5: level but
3964.5 --> 3965.3: personal
3965.3 --> 3965.9: level
3965.9 --> 3968.0: AI will
3968.0 --> 3969.0: take care
3969.0 --> 3969.8: of keeping
3969.8 --> 3970.5: the fund
3971.0 --> 3971.9: for personal
3971.9 --> 3972.5: use.
3973.0 --> 3973.4: Maybe
3973.4 --> 3973.8: actually
3973.8 --> 3974.4: shopping
3974.4 --> 3974.8: by
3974.8 --> 3975.9: coins
3975.9 --> 3976.4: that
3976.4 --> 3976.8: may be
3976.8 --> 3977.5: done
3977.5 --> 3977.9: by
3977.9 --> 3978.9: AI

3978.9 --> 3979.5: connected
3979.5 --> 3980.0: digital
3980.0 --> 3980.4: wallet.
3982.1 --> 3983.0: In
3983.0 --> 3983.4: personal
3983.4 --> 3983.7: level
3983.7 --> 3984.3: that's
3984.3 --> 3985.6: really
3985.6 --> 3986.6: a good
3986.6 --> 3987.0: future
3987.0 --> 3987.9: to come.
3988.7 --> 3989.9: That may
3989.9 --> 3990.5: be a
3990.5 --> 3991.5: valid
3991.5 --> 3992.8: future
3992.8 --> 3993.3: that we
3993.3 --> 3993.5: can
3993.5 --> 3994.0: visualize.
3995.7 --> 3996.3: And
3996.3 --> 3996.8: for the
3996.8 --> 3997.3: institutional
3997.3 --> 3998.3: level
3998.3 --> 3999.1: again
3999.1 --> 4000.9: I still
4000.9 --> 4001.9: doubt if
4001.9 --> 4002.4: that's
4002.4 --> 4002.7: kind of
4002.7 --> 4002.9: a
4004.1 --> 4005.4: if there
4005.4 --> 4005.7: is a
4005.7 --> 4006.1: way to
4006.1 --> 4006.6: explain
4006.6 --> 4007.0: that to
4007.0 --> 4007.1: a
4007.1 --> 4007.5: regulator.
4009.5 --> 4010.5: Also
4010.5 --> 4011.6: from the
4011.6 --> 4011.9: AI
4011.9 --> 4012.5: perspective
4012.5 --> 4014.9: I often
4014.9 --> 4015.1: see
4015.1 --> 4016.5: in those
4016.5 --> 4017.6: use cases
4017.6 --> 4018.5: that's
4018.5 --> 4019.3: safe to
4019.3 --> 4019.6: use.
4019.8 --> 4020.4: Sometimes

4020.4 --> 4021.4: AI goes
4021.4 --> 4022.3: very wrong
4022.3 --> 4023.1: to delete
4023.1 --> 4024.0: other databases
4024.0 --> 4025.3: or like
4025.3 --> 4026.5: I don't
4026.5 --> 4026.8: know.
4027.2 --> 4028.8: So to
4029.5 --> 4029.9: which
4030.5 --> 4030.5: extent
4030.5 --> 4031.5: like
4032.5 --> 4033.3: the AI
4033.3 --> 4034.0: is trustworthy
4034.0 --> 4034.9: or not
4034.9 --> 4035.5: that's
4035.5 --> 4035.9: really
4035.9 --> 4036.8: kind of
4036.8 --> 4037.1: a
4039.1 --> 4040.0: question
4040.0 --> 4040.5: that I
4040.5 --> 4041.4: have for
4041.4 --> 4041.9: the future
4041.9 --> 4042.7: that the
4042.7 --> 4043.3: AI is
4043.3 --> 4044.0: keeping
4044.0 --> 4044.4: the fund
4044.4 --> 4044.6: the
4044.6 --> 4044.8: right
4045.9 --> 4046.4: So
4046.4 --> 4046.4: a
4046.4 --> 4046.5: really
4048.5 --> 4048.9: question.
4048.9 --> 4049.3: I
4049.3 --> 4049.3: think
4050.2 --> 4050.7: it's
4050.7 --> 4050.9: a
4050.9 --> 4051.2: really
4051.2 --> 4051.4: it's
4051.4 --> 4051.5: a
4051.5 --> 4052.1: really
4052.1 --> 4052.2: question.
4059.1 --> 4059.6: I
4059.6 --> 4059.8: think
4059.8 --> 4060.4: it's
4060.4 --> 4060.6: a
4068.8 --> 4070.0: really
4070.0 --> 4070.1: question.

4070.1 --> 4070.1: I
4070.1 --> 4070.1: think
4076.9 --> 4078.1: the
4078.1 --> 4079.0: answer
4080.6 --> 4080.9: to
4088.5 --> 4089.7: the
4089.7 --> 4090.3: which
4090.3 --> 4090.6: is
4090.6 --> 4100.0: be more difficult to sort of give a share to an agent
or AI entity, right? So
4101.8 --> 4112.1: that may also inform the mass adoption path of this,
that MPC is for the
4112.1 --> 4119.5: institution side of things, like you use MPC solution
s, whereas for the consumer
4119.5 --> 4130.1: the individual account abstraction plus delegate auth
ority to AI makes more
4130.1 --> 4135.6: sense in terms of the journey plus security at the sa
me time.
4138.0 --> 4146.7: I've been thinking about this giving AI keys problem
a lot, so it's the solution
4146.7 --> 4152.5: that I've come up with is that you must have duality,
so you must have one that
4152.5 --> 4159.9: holds the keys and one that composes the transaction,
and neither of those systems
4159.9 --> 4168.1: can be one information system, but that's just how I
have tried to solve the
4168.1 --> 4168.5: problem.
4179.9 --> 4187.1: So it is the current state, but I think it's actually
crazy to give the access
4187.1 --> 4197.3: to your key to AI. You give the scoped key, I mean yo
u have to have AI as a
4197.3 --> 4203.5: separate process, don't run that as your own process,
right? And then you
4203.5 --> 4211.9: give a scoped key or token, right, on which the AI ca
n use it for only that
4211.9 --> 4219.8: and not beyond that thing, and you can always revoke
that. Yes, constraint, I mean
4221.3 --> 4230.9: giving, so giving out your key is just like, it's an
impersonation semantics, and
4230.9 --> 4236.8: that's very, very bad, but that's how most agents and
harnesses are
4236.8 --> 4249.1: actually operating now, and that's a big mistake. I m
ean it's no from the point of
4249.1 --> 4254.1: view of the identity management. That actually brings
a kind of our discussion
4254.1 --> 4258.4: like topic back to fund management as well, like well
kind of our separation of

4258.4 --> 4262.9: wallet, like kind of large wallet into like smaller pieces of wallet. For

4262.9 --> 4268.6: example, like there is 1 million USD worth Bitcoin wallet here, but like

4268.6 --> 4274.2: there is only a thousand USD worth Bitcoin wallet in here. Maybe like

4274.2 --> 4283.3: those kind of small wallet key can be like delegated to, not AI, but I

4283.3 --> 4288.3: don't know, like some kind of automated management agent or systems anyway. So

4288.3 --> 4294.3: that's totally kind of valid way of managing the fund as well. Then like well

4294.3 --> 4301.6: what would be the kind of right way to, you know, assess the risk, I mean like what

4301.6 --> 4308.0: would be the kind of evaluation of like how much wallet can be managed by what

4308.0 --> 4313.5: kind of a system, like autonomous system. That's kind of a similar question when

4313.5 --> 4320.8: the DAO came up as well. So and I think that's really have no answer, but it's

4320.8 --> 4324.7: really kind of interesting like sort of experiment as well.

4329.2 --> 4336.4: I'm curious about user experience. So you know we are talking about technical

4336.4 --> 4344.5: stuff as usual, but you know operator and executives has not like us. They don't

4344.5 --> 4351.8: have enough literacy, I guess, right. So as we're talking about what is a secure

4351.8 --> 4357.7: system, but they don't need any other kind of stuff. So I guess we need a

4357.7 --> 4367.1: retrain them all, right. So we can imagine the kill chain, but at the first

4367.1 --> 4375.9: step, adversaries will investigate operators, but if they don't have enough

4375.9 --> 4386.1: literacy, they collapse. So I think we should imagine how the operators behave

4386.1 --> 4395.4: and what they think about it, right. Can you give me something like your personal

4395.4 --> 4396.4: opinions?

4400.3 --> 4409.7: I mean that's really kind of essential part, I mean like whether literacy of

4409.7 --> 4414.8: like operating key, of course. So like well from the custodian perspective, like

4414.8 --> 4420.1: operators or system designer or I don't know like operational company themselves.

4420.9 --> 4425.1: Anyway like what those people should be professional, because like what they are

4426.5 --> 4432.6: keeping the fund from the customers, like as a custodian. So they should know what

4432.6 --> 4438.4: the risk and they should know the right way, I mean proper way to operate the key

4438.4 --> 4445.5: of course. So in such a sense, like what those people should be knowledgeable

4445.5 --> 4450.6: about what would be their potential risk, what would be the like potential

4450.6 --> 4460.5: adversarial risk as well. And in that sense, like in broader cybersecurity

4461.2 --> 4466.9: context, those people like employees or like members in those kind of company

4466.9 --> 4471.1: should be properly educated as well, like well from the cybersecurity perspective.

4471.5 --> 4481.9: And that's really essential and I mean like well that's like I mean you know

4482.8 --> 4488.0: kind of foundational, I mean not really different from the current state of their

4488.0 --> 4493.5: exchange operation as well. And of course like was a regular like normal users

4493.5 --> 4499.7: like well I mean like kind of end-users perspective, they don't have proper

4499.7 --> 4506.7: knowledge or they don't have, they may not have any kind of information like IT

4506.7 --> 4511.1: knowledge as well. Like what we as Japanese people like have my number card

4511.1 --> 4519.4: right? And those my number card have a key to your like Jitsuin, like equivalent

4519.4 --> 4524.8: as well. And that's really like important key for like people, I mean

4524.8 --> 4529.9: citizens key as well. But most of the people use like well their passcode as

4529.9 --> 4535.3: their birthday or anything. So that's kind of literacy that we we may be able

4535.3 --> 4539.8: to expect from their like regular users. So that's kind of a different

4540.6 --> 4545.4: expectational level for end-users and institutional level, I think.

4550.4 --> 4554.4: We have only five minutes but yeah.

4556.6 --> 4560.1: Originally, so I mentioned at the beginning of this session, I mentioned

4560.1 --> 4564.9: that it's a kind of original you know background to start that is the

4564.9 --> 4569.6: competition for key management is just a standardizing MPC or

4569.6 --> 4573.4: something like that. It's a kind of a replacement of that, you know, but

4575.8 --> 4581.3: practice to that standardized practice throughout, though the you know

4581.3 --> 4588.7: competition process. But so you know that a great brainstorming about that. So how

4588.7 --> 4593.4: we create that more innovative and better key management systems for

4593.4 --> 4599.4: crypto assets. Like potentially it might be a replacement of the old fashioned

4599.4 --> 4605.5: cold, hot, something like that. Or introducing some AI or something like

4605.5 --> 4612.0: that. Anyway that to the to make the key management system more manageable from

4612.0 --> 4617.9: the risk management point of view or introducing AI to reduce

4618.9 --> 4623.1: points of the human operation or something like that. Anyway it's a

4623.1 --> 4629.0: risk management point of view. But so we can be a more

4629.0 --> 4634.3: innovative to if it's a three years competition. So we cannot imagine what

4634.3 --> 4639.2: happening three years after now. But so we can be more innovative to create

4639.2 --> 4645.0: better key management system for crypto assets. So that is what we should do.

4645.7 --> 4650.6: It's not just a replacement of the old thing. And so we need to discuss

4650.6 --> 4658.1: with about it with you just about it. But so it's a based on that the Japanese

4658.1 --> 4665.7: academics. And we should be that more innovative. And so we that this

4665.7 --> 4670.0: topic, key management topic is also the topic we discussed in the October

4670.0 --> 4677.0: Computer Security Symposium. And yeah let's design that session.

4678.2 --> 4682.6: You know that not only on that very old fashioned key management discussion.

4682.7 --> 4687.5: But how we can make more innovative including that introducing AI

4687.5 --> 4692.8: something like that. That is a key take away from this session. It's a great

4692.8 --> 4698.9: great suggestion. So the original idea is just a replacement. But we can do more.

4710.6 --> 4717.4: Okay so I said just to make announcement. So we have one hour lunch break after

4717.4 --> 4725.4: that at 1.30. So a keynote by JFS 8 followed by that two sessions.

4727.5 --> 4734.5: It's just only Japanese audience. But we have the fol

low-up session at 5.30 or

4734.5 --> 4738.7: 5.10 for Japanese audience to have some wrap-up.

4741.4 --> 4746.2: Yeah but anyway that's so we have two more sessions in the afternoon. Two plus

4746.2 --> 4750.4: one more sessions in the afternoon. So please join the session. Thank you very

4750.4 --> 4754.9: much and have a great lunch and come back here at 1.30 p.m.

4757.3 --> 4760.7: Thank you. Thank you everyone for the remote participant including Andy.

4780.1 --> 4789.2: We will start that the afternoon session by the keynotes by JFS 8 and it will be

4789.2 --> 4796.8: happened in room 8, the other room. After that we have the session, two sessions at

4796.8 --> 4800.2: 1.50 p.m. Japanese time.

4834.8 --> 4836.2: Thank you.